

Applied AI Engineering Intern — Self-Check Loop Write-up

Project: AI Medical Report Extraction Self-Checker

Failure targeted: fabricated or incorrect extracted facts, especially plausible wrong numeric lab values.

1. Problem selected

The tool extracts a small set of laboratory values from a medical-report text. The chosen reliability failure is a plausible numeric value that is incorrect but looks believable.

2. Two-pass design

Pass 1 creates structured extraction. Pass 2 independently obtains source-supported values and compares every draft value. MATCH values are accepted, mismatches are corrected from the source, and unsupported fields are flagged.

3. Confidently wrong example

Source: Total Cholesterol: 185 mg/dL. First draft: Total Cholesterol: 285 mg/dL. The second pass detects the mismatch and corrects the final value to 185 mg/dL.

4. Five test cases

Test 1: correct extraction; Test 2: plausible wrong cholesterol; Test 3: wrong hemoglobin; Test 4: unsupported field; Test 5: missed glucose field. The tests cover both clean and failure paths.

5. Results

The repository contains five automated tests. Expected local run: 5 passed.

6. Where the system failed

A first-pass system can produce a fluent, plausible number that is not grounded in the source. Returning that answer directly would hide the failure.

7. What I did about it

I made the first pass untrusted and added source-grounded verification. The checker compares each value, identifies mismatches, corrects them using source evidence, and reports what it found.

8. Limitations

This is a technical demonstration, not a clinical system. It uses text rather than scanned documents, supports a small field set, does not diagnose conditions, and requires human/clinical validation for any real deployment.

9. Key lesson

A confident first answer is still a draft. A useful reliability loop needs a specific, observable property to verify. Here, every extracted value must be supported by the source text.

Before → After evidence

SOURCE: Total Cholesterol: 185 mg/dL
DRAFT: Total Cholesterol: 285 mg/dL
CHECK: MISMATCH
FINAL: Total Cholesterol: 185 mg/dL

Repository contents

app.py
extractor.py
checker.py
requirements.txt
README.md
tests/
results/
docs/